

IS883: Deploying Generative AI

Mohannad Elhamod



Model Access and Using APIs

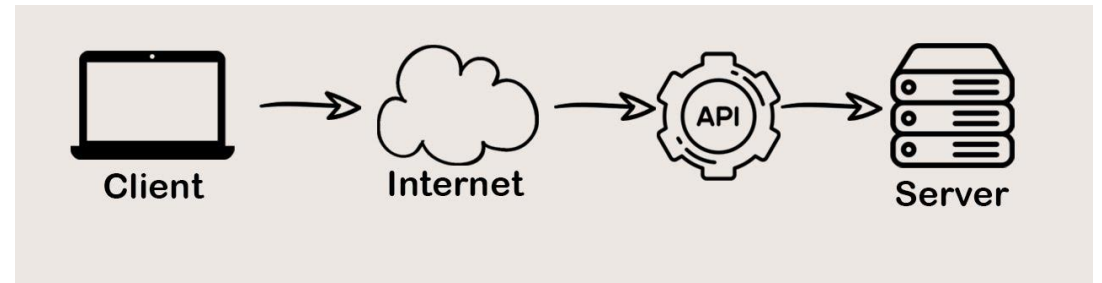
Model Access

- **Open source (Rare):** You have model parameters, training code, and data information.
- **Open weights:** You download the parameters and run them anywhere. The training data and the recipe stay private, and the licence is the vendor's own.
 - Llama 3.2 / Llama 4 Scout, gpt-oss, Qwen, Mistral, DeepSeek.
- **Closed** — you never touch the weights. You send a prompt and rent the answer (Basically through an API).
 - Gemini, Claude, GPT-4o.

What is an API?

An Application Programming Interface (API) establishes a “contract” or “glue” that allows multiple pieces of software to communicate.

- Standardizes software development.
- Allows integration with online services.
- Is generally well-documented.



[CodeWithMazn](#)

LLM APIs

APIs that provide access to LLMs:

- You send a query/request (e.g., a prompt or a sentence)
- You get a response (e.g., a summary or a translation)



Why LLM APIs (vs. Hosting)?

- Pros?
 - Less memory and compute resources needed.
 - Abstraction in terms of maintenance and updates.
 - Security?
- Cons?
 - Less control.
 - Maybe slower.
 - Cost.
 - Privacy.
 - Vendor risk.



Host or Rent?

- Open-weight models are more open to interpretability.
- Two paths, one identical request:
 - Hosting (e.g., [Llama 3.2 3B](#) on a Colab GPU): No per-token charge, works offline, capped by your hardware.
 - Renting: (e.g., [Llama 4 Scout](#) through the HuggingFace API): $\sim 30 \times$ larger, nothing downloaded, billed per token.
- At our volume, renting is the **cheaper** path. Privacy, vendor risk and volume are why companies self-host — not cost.

Gemini API

- [API Documentation](#)
- [Playground](#)
- [General Documentation](#)

Cloud Computing

What is Cloud Computing?

The technology that allows us to access computing resources over the internet (as opposed to purchasing our own resources).



Google Cloud



Subscription vs. Buying



[wearekemb](https://www.wearekemb.com/)



Google Cloud



StreamLit for Web App Hosting

- Will allow us to create a user interface for our application
- [Creating a dummy app](#)
- [A StreamLit with Gemini Prototype](#)



When the API Says No

- The free tier (and some times the paid one) caps how many requests you may send per minute. You may hit that in your project, so plan accordingly.
- Our **ask_gemini** helper already catches it. “Waiting 15 seconds...” means it is retrying for you — let it. If it keeps failing: wait, send fewer and shorter requests, then come back to it.

Model Size

- **Parameters in billions $\times$ 2 bytes per parameter = GB of memory.** \
- Streamlit Community Cloud gives you **1 GB and no GPU**.
 - Llama 3.2 3B $\rightarrow 3 \times 2 =$ **6 GB needed**. Can it load?
 - Qwen2.5-0.5B $\rightarrow 0.5 \times 2 =$ **1 GB**. Can it load?
- The weights are the minimum needed. More memory is needed for the code, prompts, software, etc.
- You could do this arithmetic before you deploy. The failure is only confirmation.

Code Development Tools

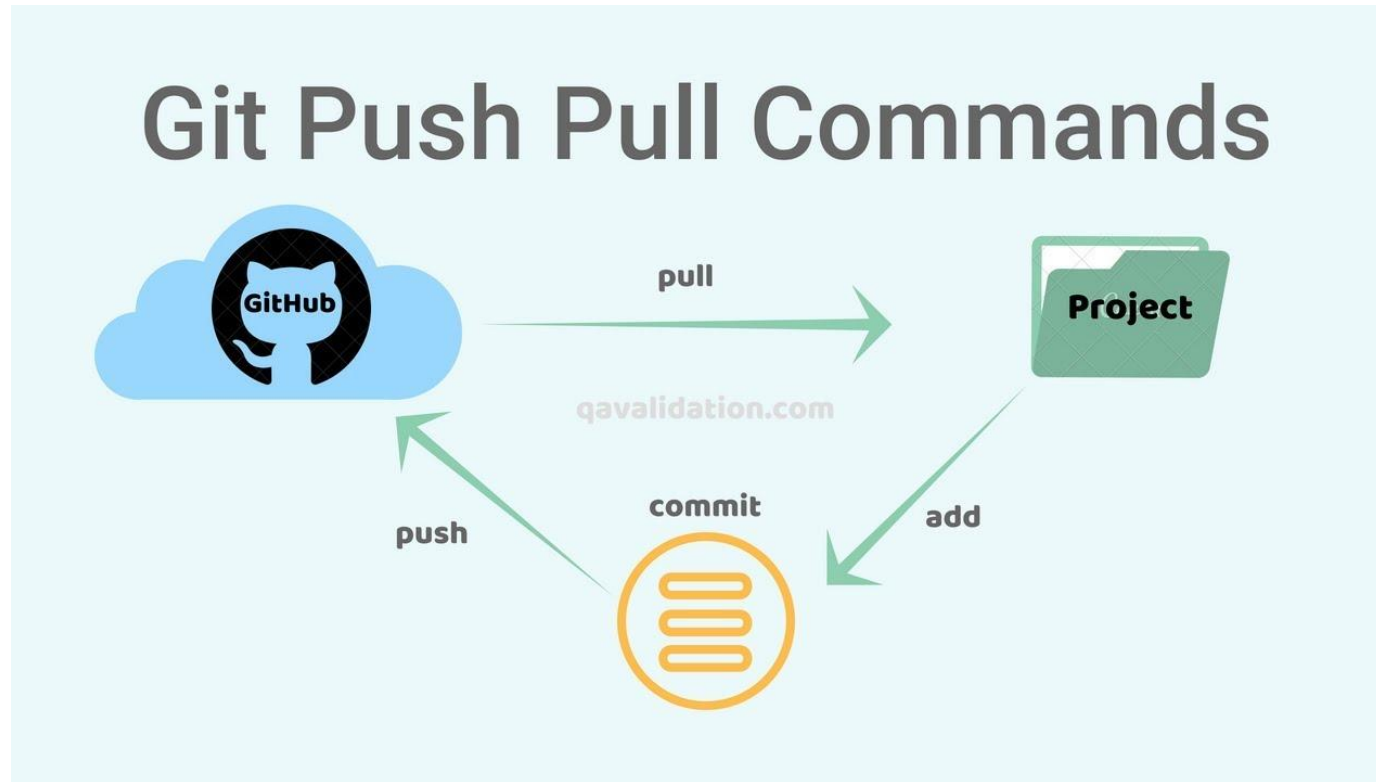
GitHub

Widely used.

- Helps with team collaborations.
- Tracks code change history.
- Can automate tasks such as deployment.
- Allows multiple versions of the code (e.g., for different features).



The Git Workflow



VS Code

A powerful editor by Microsoft

- Lightweight and easy to use.
- Can integrate with GitHub.
- Helpful features: IntelliSense.



In-Class Work